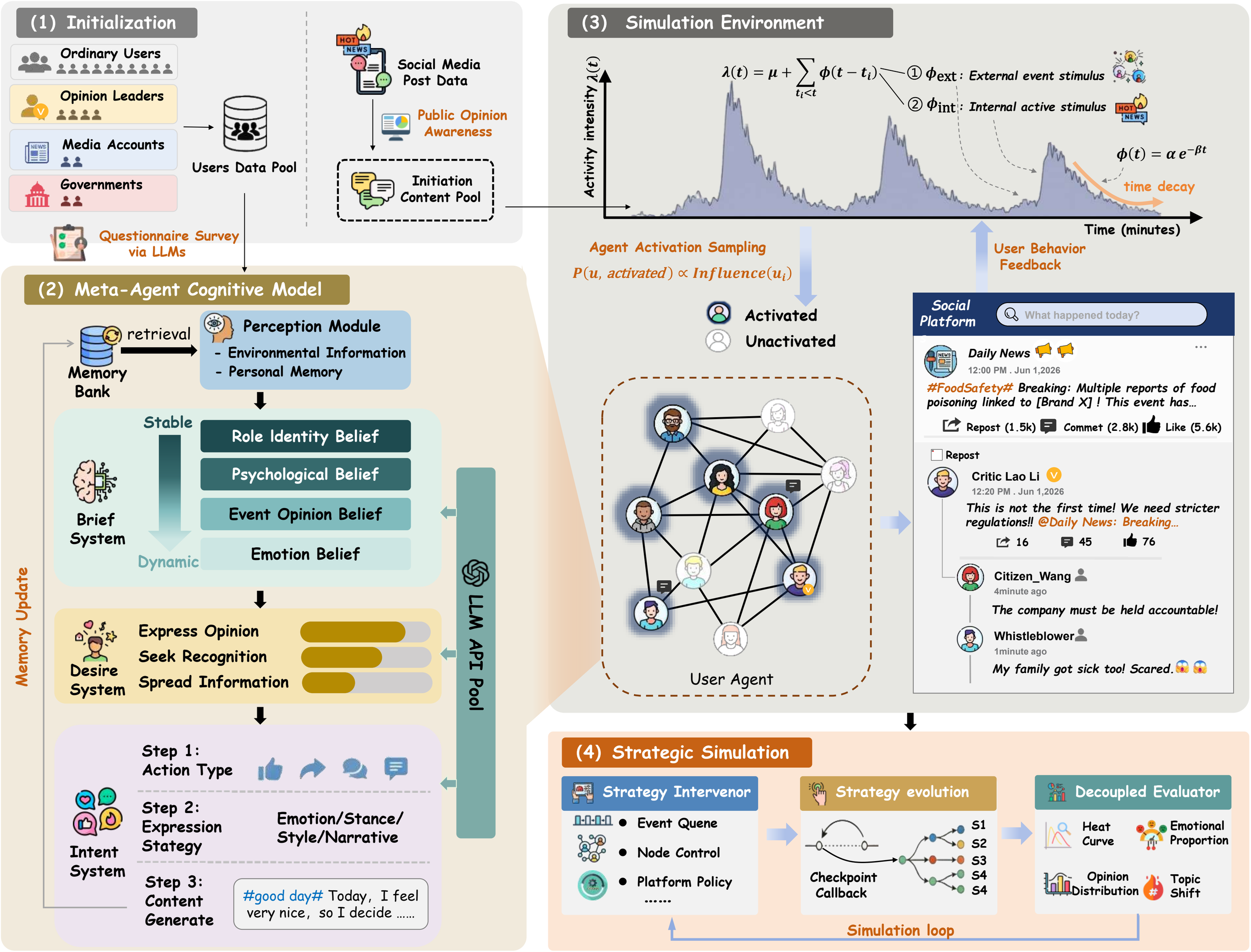
(1) Initialization
Ordinary Users
Opinion Leaders
Media Accounts
Governments
Users Data Pool
Social Media Post Data
Public Opinion Awareness
Initiation Content Pool
Questionnaire Survey via LLMs
(2) Meta-Agent Cognitive Model
Memory Bank
retrieval
Perception Module
- Environmental Information
- Personal Memory
Stable
Dynamic
Brief System
Role Identity Belief
Psychological Belief
Event Opinion Belief
Emotion Belief
Desire System
Express Opinion
Seek Recognition
Spread Information
Intent System
Step 1: Action Type
Step 2: Expression Stategy
Emotion/Stance/ Style/Narrative
Step 3: Content Generate
#good day# Today, I feel very nice, so I decide .....
LLM API Pool
Memory Update
(3) Simulation Environment
Activity intensity λ(t)
λ(t) = μ + Σ φ(t − t_i)
① φ_ext : External event stimulus
② φ_int : Internal active stimulus
φ(t) = α e^{−βt}
time decay
Time (minutes)
Agent Activation Sampling
P(u, activated) ∝ Influence(u_i)
User Behavior Feedback
Activated
Unactivated
User Agent
Social Platform
What happened today?
Daily News
12:00 PM . Jun 1,2026
#FoodSafety# Breaking: Multiple reports of food poisoning linked to [Brand X] ! This event has...
Repost (1.5k) Commet (2.8k) Like (5.6k)
Repost
Critic Lao Li
12:20 PM . Jun 1,2026
This is not the first time! We need stricter regulations!! @Daily News: Breaking...
16 45 76
Citizen_Wang
4minute ago
The company must be held accountable!
Whistleblower
1minute ago
My family got sick too! Scared.
(4) Strategic Simulation
Strategy Intervenor
Event Quene
Node Control
Platform Policy
.....
Strategy evolution
Checkpoint Callback
S1
S2
S3
S4
S4
Decoupled Evaluator
Heat Curve
Emotional Proportion
Opinion Distribution
Topic Shift
Simulation loop

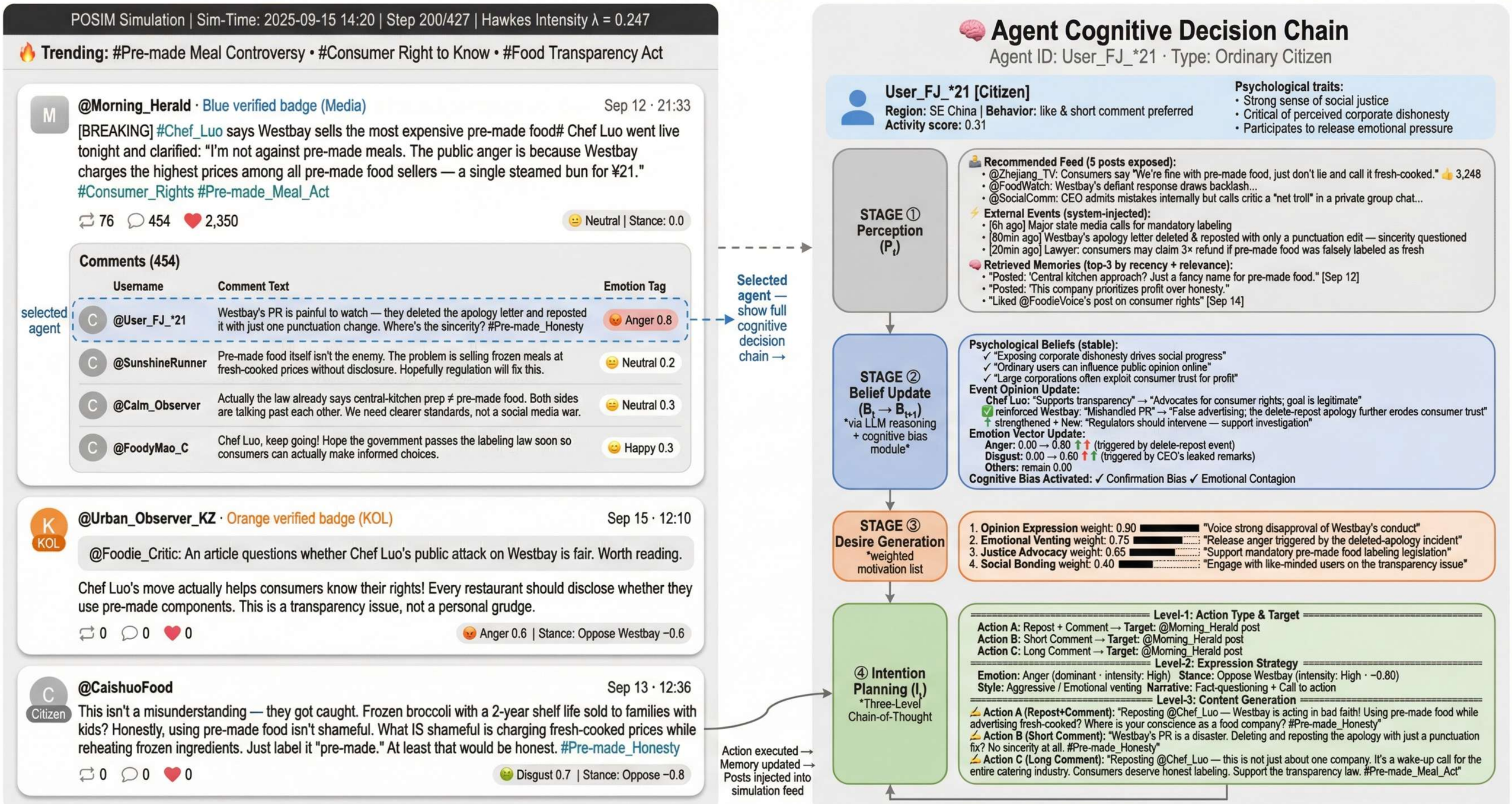

Figure based on real POSIM simulation data (Step 200, Sim-Time: 2025-09-15 14:20). This step: Hawkes process activated 43 agents [citizen: 24, KOL: 6, media: 12, gov: 1], generating 135 actions (post: 32, repost: 35, comment: 65, like: 3). Agent identifiers are anonymized for privacy. All post content is generated by LLM-driven agents.